

Name: [REDACTED]

Date: [REDACTED]

Total
29
/30

Knowledge

1. Which pair of vectors are not collinear? (Circle One)

a) $2\vec{a}, -5\vec{a}$

b) $\pi\vec{v}, 7.38\vec{v}$

c) $2\vec{b}, 2\vec{a}$

d) $\vec{c}, -\vec{c}$

(1)

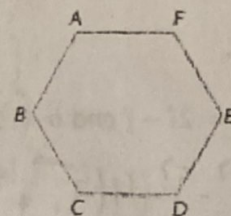
2. Given the diagram to the right, which statement is true? (Circle One)

a) $\vec{AF} = \vec{DC}$

b) $\vec{CD} + \vec{DE} = \vec{CF}$

c) $|\vec{FC}| = |\vec{BE}|$

d) $\vec{AF} + \vec{DE} = \vec{BA}$



(1)

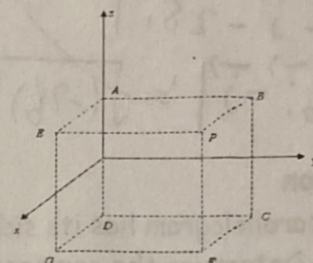
3. The prism shown is bisected by the xy -plane. The point $P = (3, 7, 3)$. Determine C .

a) $(0, 7, 3)$

b) $(0, -7, 3)$

c) $(0, -7, -3)$

d) $(0, 7, -3)$



(1)

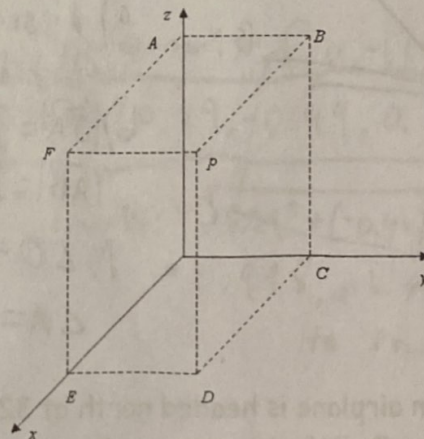
4. The point $P = (8, 5, 7)$. Determine the equation for the plane containing points B, C, D , and P .

a) $y = 5$

b) $x = 5$

c) $z = 5$

d) $x = 8$



(1)

5. Determine vector $\vec{AD}$ if $A(5, 8, -7)$ and $D(4, -3, -16)$.

$\vec{AD} = (4-5, -3-8, -16+7) = (-1, -11, -9)$

$(-1, -11, -9)$

(1)

6. Given $\vec{x} = (1, -3, 2)$, $\vec{y} = (-12, 6, -9)$, and $\vec{z} = (7, -10, 4)$, determine a vector for $2\vec{x} + \frac{1}{3}\vec{y} - \vec{z}$.

$$\begin{aligned} & 2(1, -3, 2) + \frac{1}{3}(-12, 6, -9) - (7, -10, 4) \\ &= (2, -6, 4) + (-4, 2, -3) + (-7, 10, -4) \\ &= (-9, 6, -3) \end{aligned}$$

(2)

$$(-9, 6, -3)$$

7. If $\vec{a} = 2\vec{i} - \vec{j}$ and $\vec{b} = 3\vec{i} + 2\vec{j}$, determine $|\vec{a} - 4(\vec{b} + 2\vec{a})|$. Round to 2 decimal places.

$$\begin{aligned} &= |2\vec{i} - \vec{j} - 4(3\vec{i} + 2\vec{j} + 4\vec{i} - 2\vec{j})| = \\ &= |2\vec{i} - \vec{j} - 4(7\vec{i})| = \\ &= |2\vec{i} - \vec{j} - 28\vec{i}| = \\ &= |-26\vec{i} - \vec{j}| = \sqrt{(-26)^2 + (-1)^2} = 26.02 \end{aligned}$$

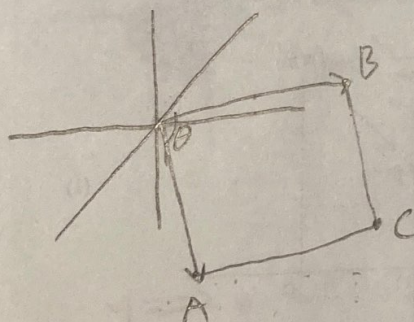
(3)

$$26.02$$

Application

8. A parallelogram has its sides determined by the vectors $\vec{OA} = (3, 2, -6)$ and $\vec{OB} = (-6, 6, -2)$.

- Determine the components of the vectors representing the diagonals.
- Determine the angles between the sides of the parallelogram.



a) diagonal $= \vec{AB} = (-6-3, 6-2, -2+6) = (-9, 4, 4)$ (4)

other diagonal is perpendicular, therefore it $= (9, -4, -4)$ ~~$(-3, 9, -8)$~~

b) $|\vec{OA}| = \sqrt{3^2 + 2^2 + (-6)^2} = 7$ $|\vec{OB}| = \sqrt{(-6)^2 + 6^2 + (-2)^2} = 8.72$

$|\vec{AB}| = \sqrt{(-9)^2 + 4^2 + 4^2} = 10.63$

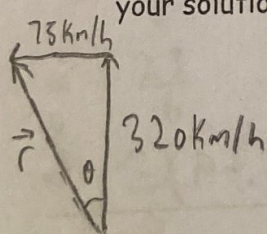
$\angle O = \angle C = 84.34^\circ$

$\angle A = \angle B = 95.66^\circ$

use cosine law
 $(10.63)^2 = 7^2 + 8.72^2 - 2(7)(8.72)\cos\theta$
 $-12.0415 = -122.08\cos\theta$
 $\theta = 84.34$

$\frac{1}{2}(360 - 84.34(2)) = 95.66$

9. An airplane is headed north at 320 km/h. The plane encounters a wind from the east blowing at 75 km/h. What is the resulting ground velocity of the plane? Include a vector diagram as part of your solution.



$r^2 = 75^2 + 320^2$

$r = \sqrt{108025} = 328.67 \text{ km/h}$

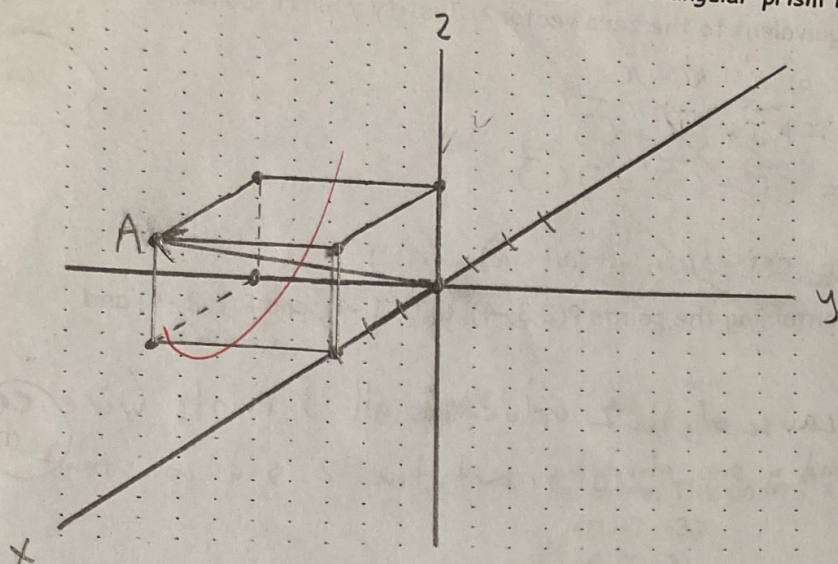
$\tan\theta = \frac{75}{320}$

$\theta = 13.19$

$\therefore 328.67 \text{ km/h } [N 13.19^\circ W]$

(3)

10. Locate the vector $\vec{OA} = (3, -5, 4)$ using the rectangular prism method. Label the axes.



- $(0, 0, 0)$
- $(3, 0, 0)$
- $(0, -5, 0)$
- $(3, -5, 0)$
- $(0, 0, 4)$
- $(3, 0, 4)$
- $(0, -5, 4)$
- $(3, -5, 4)$

(2)

11. $V(2, 8, -15)$ and $U(9, -4, 5)$ represent two points in R^3 .

- a) Determine the components of vector $\vec{VU}$.
- b) Determine the components of the unit vector in the same direction as $\vec{VU}$.

a) $\vec{VU} = (9-2, -4-8, 5+15) = (7, -12, 20)$

b) $|\vec{VU}| = \sqrt{(7)^2 + (-12)^2 + (20)^2} = \sqrt{593}$

$1 = 49a^2 + 144a^2 + 400a^2$

$1 = 593a^2$

$a^2 = \frac{1}{593} \Rightarrow a = \pm \frac{1}{\sqrt{593}} = \pm \frac{1}{24.35}$

take only positive answer to keep same direction

$\therefore$ components: $(7a, -12a, 20a)$
 $= \text{about } (0.29, -0.49, 0.82)$

checks: $\sqrt{0.29^2 + (-0.49)^2 + 0.82^2} = 0.998$, a bit off due to rounding

Communication

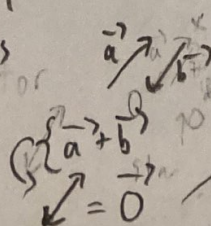
12. If two vectors are opposite, is their sum a vector? Explain with a use of a diagram.

Yes, their sum will be a vector.

If 2 opposite vectors

1. are added, they will cancel out and give $\vec{0}$, which is still a vector.

2. same mag. as well as opp. direction



while mag. is 0, it is still a 0 vector

(2)

13. Which expression(s), if any, is equivalent to the zero vector? Justify your response.

a) $\vec{EU} - \vec{EP} - \vec{PU}$
 $= \vec{EU} + \vec{PE} + \vec{UP}$
 $= \vec{EU} + \vec{UE}$
 $= \vec{EU} - \vec{EU} = \vec{0}$

b) $\vec{KJ} - \vec{KC} - \vec{JC}$
 $= \vec{KJ} + \vec{CK} + \vec{CJ}$
 $= \vec{CJ} + \vec{CJ} = 2\vec{CJ}$

$\therefore$ expression in part a) only

(2)

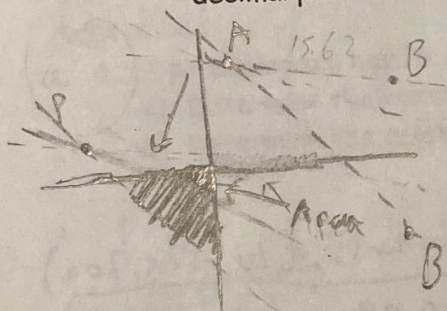
14. State the equation of the plane containing the points $P(2,3,-4)$, $Q(0,-1,-4)$, and $R(6,2,-4)$ and describe how you know.

$z = -4$, I know this because only the z value in all 3 points were equal, this means that the x and y are changing, but the z is a constant

(1)

Thinking

14. Points $A(1, 6)$ and $B(13, -4)$ lie in $\mathbb{R}^2$. Determine the enclosed area a line that is parallel with $\vec{AB}$ creates with the x and y axis, if this line goes through the point $P(-7, 2)$. Round to 2 decimal places.



$\vec{AB} = (12, -10)$

$|\vec{AB}| = \sqrt{12^2 + (-10)^2} = 15.62$ though parallel doesn't mean same length

(3)

slope

$m = \frac{-4-6}{13-1} = \frac{-10}{12} = \frac{-5}{6}$

$y = \frac{-5}{6}x + b$

use point P

$2 = \frac{-5}{6}(-7) + b$

$\frac{-23}{6} = b$

$y = \frac{-5}{6}x - \frac{23}{6}$

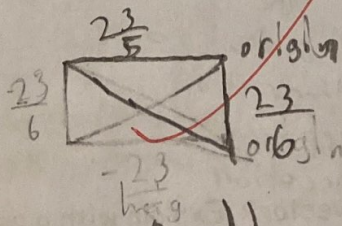
$y = \frac{-5}{6}x - \frac{23}{6}$

find intercepts

$y = -\frac{23}{6}$ $(0, -\frac{23}{6})$

$-\frac{5}{6}x = \frac{23}{6}$

$x = -\frac{23}{5}$ $(-\frac{23}{5}, 0)$



$A = \frac{bh}{2} = \frac{\frac{23}{5} \cdot \frac{23}{6}}{2} = \frac{529}{60} = 8.82 \text{ units}^2$